ELECTRICAL ENGINEERING:
Possibilities and Limitations - ETI 250

Lecture no: **12**

Datorers kringutrustning

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Innehåll

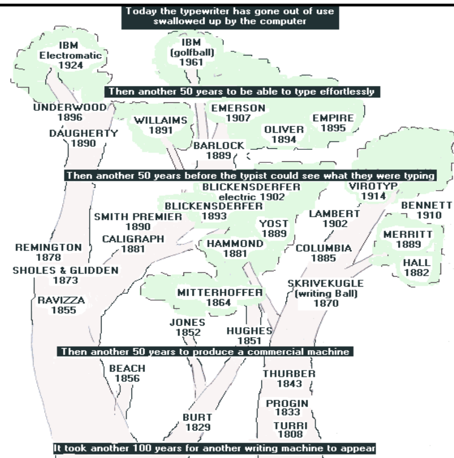
- Att ge data till datorn
  - tangentbord
  - möss
  - track balls
  - Scanners
  - joy sticks/game control
- ...och att "presentera" resultat
  - skivare
  - skärmar beskrivit tidigare

Keyboards

The original ?

In 1865 Rev. Rasmus Malling-Hansen of Denmark invented the Hansen Writing Ball... the first typewriter to allow the operator to write substantially faster than a person.
Wikipedia

Typewriter......produced by the gunmakers E. Remington & Sons from 1874-1878. ...not more than 5,000 sold...



In 1714, the last year of her reign, Queen Anne granted a patent to Henry Mill.

It was patent no 385 and Henry Mill affirmed that he had:

"..by his great study, pains and expense, lately invented and bought to perfection an artificial machine or method for the impressing or transcribing of letters, singly or progressively one after the other, as in writing, whereby all writings whatsoever may be engrossed on paper or parchment so neat and exact as not to be distinguished from print .. the impression bing deeper and more lasting than any other writing, and not to be erased or counterfeited without manifest discovery". Details of Henry Mill's machine are not known.

...and the followers

The mechanical problems...

and finally the electronic once...

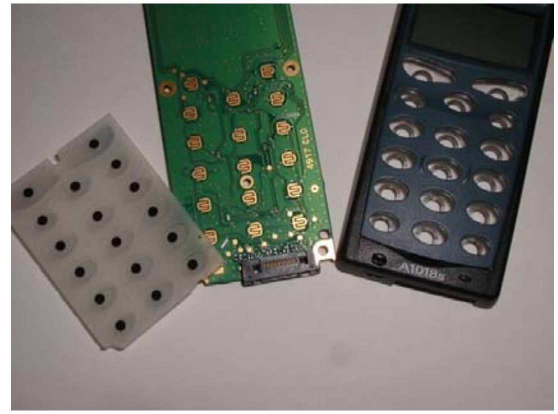
Electronic typewriter - the final stage in typewriters development.
A 1989 Canon Typestar 110

Inside the keyboard

Keyboards use a variety of switch technologies. It is interesting to note that we generally like to have some audible and tactile response to our typing on a keyboard. We want to hear the keys "click" as we type, and we want the keys to feel firm and spring back quickly as we press them.

- Rubber dome mechanical
- Capacitive non-mechanical
- Metal contact mechanical
- Membrane mechanical
- Foam element mechanical

Inside the keyboard, another example



Virtual Keyboard

The lifting and lowering of the users finger at a virtual key's location indicates the intention of a user to "strike that particular key." Such intention and position can be determined remotely using laser triangulation and miniature cameras. The angle of the reflected laser light off the finger determines the position (hence the "which key") context. The required components including a laser and two miniature CCD cameras...

Wireless solutions...

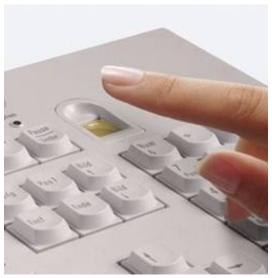
e.g. 27Mhz, Bluetooth,...

+ no cables

- connectivity, battery lifetime, ...?

Security

Security: fingerprint, smartcard,...



The Mouse

Types: One/Two/Three ... button
”Ball”, optical, optical+”ball”,...

Interfaces: Serial Port, PS/2 Port, USB, etc.

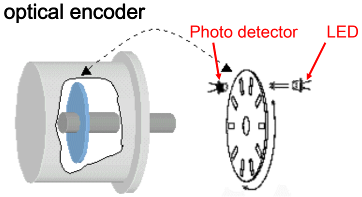
Mechanical mouse - principle

Image from www.wikipedia.org

1. Ball movement
2. Rollers picking up ball movement
3. Optical encoding disks
4. Infrared LEDs
5. Photo sensors

Sensing rotation

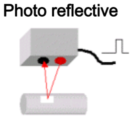
Incremental optical encoder



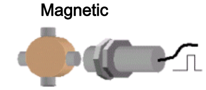
One pulse from the photo detector for each slit in the rotating disc.

Some other types:

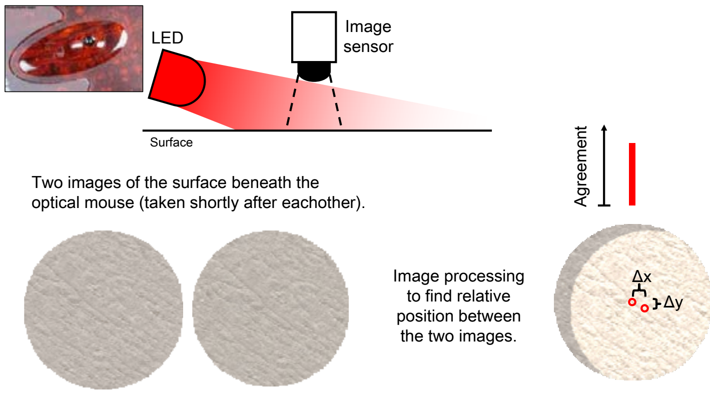
Photo reflective



Magnetic



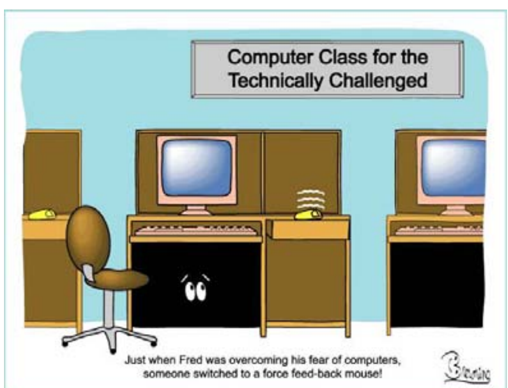
Optical mouse - principle



Optical mouse - performance

- Parameters determining performance
  - Size of image sensor
    - Larger sensors get more information about the surface
  - Refresh rate (number of surface images per second)
    - The faster the refresh rate, the better the tracking performance
  - Image processing capacity
    - Faster image processing can improve both precision and tracking speed

Mouse ... next version?



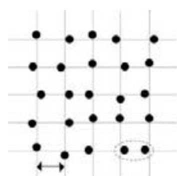
Track ball

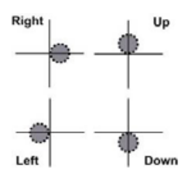
Three dimensions ... with force feedback

”Electronic pen” (Anoto)

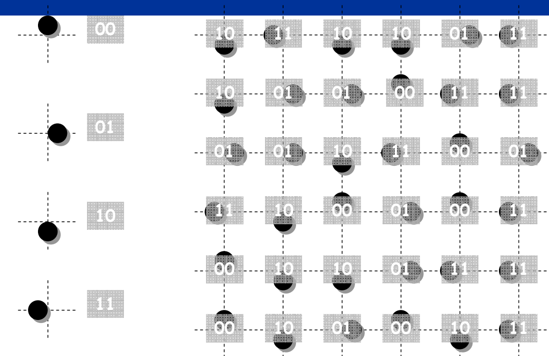
The Anoto pattern

- Unique Proprietary Pattern
- Determines Pen’s Position
- Printable on Most Surfaces
- Pattern Linked to Specific Functions
- 6e14 unique pages





Decoding



$z =$ 101110100111100101001111010110110001111000010011001010011111001001001011

$$f(z) \rightarrow (x, y)$$

Tablet

Usually connects to PC through USB

Bar-code scanner

Laser

CCD – close

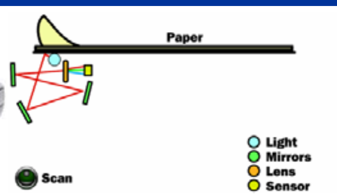
Wand

Wireless

Slot

Image scanner

Flatbed



Sheetfed

Web and Network cameras

From small, poor quality, PC connected cameras

to

Larger, high quality, stand-alone network cameras

Is connected to a PC through USB.

Is connected cirectly to the computer network through EtherNet.

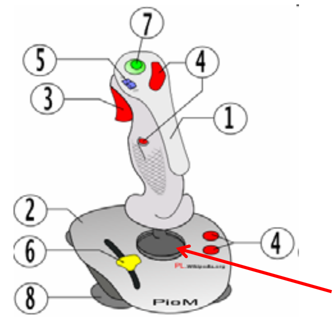
Paddle

A very simple type of game controller is a so called paddle, with a **wheel** and optionally one or more **buttons**.

Turning the wheels changed the resistance of a potentiometer

The classic PONG game

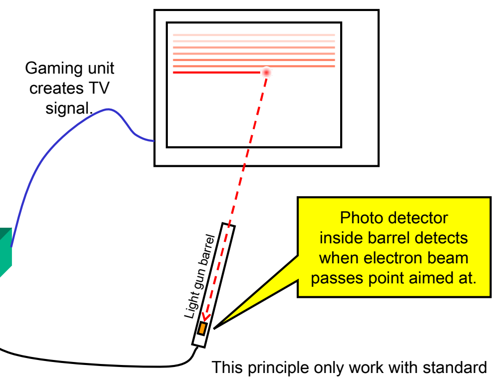
Joystick



ANALOG: Stick connected to two potentiometers sensing tilt in two directions.

DIGITAL: Stick connected to switches giving on/off signals in four different directions (up/down/left/right).

Light gun – cathode ray timing



This principle only work with standard cathode ray tube monitors (TVs).

Light gun – infrared emitters

Since cathode ray timing only works with standard CRT monitors, an alternative is needed for 100 Hz CRT- , LCD- , Plasma- and DLP displays.

IR camera close to the display

Light gun with IR transmitters

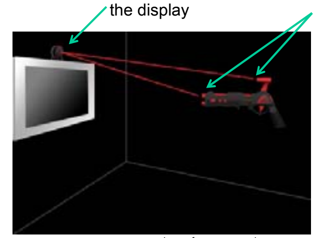


image from www.erealgames.com

After calibrating the camera, the system can calculate where on the display the light gun is pointing. (Alternative systems exist with IR transmitters on the display and sensors in the gun.)

Steering wheel, with force feedback

Motor creates feedback force in the steering column.

Some other game controllers

Dance pad with switces in a pastic mat.

Output devices

Display

Printers

Bubble/Ink jet

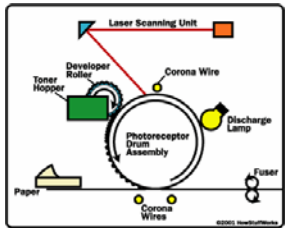
small drops of ink applied to surface

Technology developed at LTH by Hellmuth Hertz

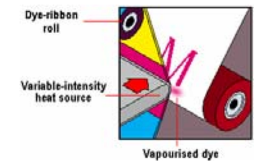
Laser

- Drum is charged by corona wire
- Laser discharge according to pattern
- Toner is attracted/repelled
- Toner put on paper by second corona wires(s)

- color – several steps for the different colors



Dye-sublimation printer



Produces (expensive) photo-realistic continuous-tone images that look like they came from a photo lab.

Usually connects to PC through USB or parallel port (printer port) and also prints directly from memory cards.

Thermal paper printer

Heated elements in contact with special heat sensitive paper forms darkened dots when the elements reach a critical temperature.

Mostly for special applications, like cash registers, fax machines, etc.

Print quality is generally low and the paper darken over time due to exposure to sunlight and heat.